

A Comparative Analysis of VR-Based and Real-World Human-Robot Collaboration for Small-Scale Joining

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ABSTRACT

While VR as an interface for the teleoperation of robots has been well-studied in recent years, VR can also be used to advance our understanding of in-person human-robot interaction (HRI) by simulating such interactions more repeatably and affordably than real-world studies. A few platforms now exist for studying human-robot interaction in VR, but little of this work has involved the study of the realism of specific, typical in-person HRI tasks. To evaluate this realism, we conduct a user study consisting of a collaborative assembly task where a robot and human work together to build a simple electrical circuit. We present a comparison of the task performed in the real world versus with virtual robot in VR. We discuss difficulties encountered and draw conclusions about what characteristics a virtual environment should have in order to support physical human-robot interactions.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Collaborative interaction;

1 INTRODUCTION

As robots become less expensive and easier to obtain, human-robot interactions (HRI) will increasingly take place in less controlled environments. Rather than being confined to factory floors and warehouse work, robots will become more engaged in collaborating with the people on common human tasks. Such tasks require attention to detail to avoid harming human collaborators or damaging manipulated objects. Virtual reality provides an opportunity for developing and testing approaches to these human-robot interactions, with immersed humans and simulated robots acting as stand-ins for their real-world analogs [5]. However, a VR simulation is not guaranteed to be a reliable surrogate for how humans, robots, and environments behave in the real world. This work is focused on exploring these differences in an HRI task, both by reviewing the experiences of study participants and through our observations of the system as a whole.

Although there has been extensive work on sim2real in robotics [2], most of the work in human-robot interaction studies has focused on the experience of the person in the interaction, and less on actual, physical collaborations or interactions between a person and a robot. For this work, we conducted an experiment on small-scale joining of objects in which the human and robot worked together on a manipulation task. The experiment was run both in a virtual reality environment and on a physical robot, and we compare the successes and failures of the experiment in both cases, examining when behaviors are specific to virtual reality as opposed to being correctly shared across platforms.

Unlike previous work on collaboration in industrial settings [1, 3, 4] and on larger scale assembly tasks [6, 7], we focus on small-scale

tasks relevant to daily life. As an exemplar, we consider the specific example of shared manipulation for joining two objects together. In this task, a robot serves as a human's "third hand." A variety of joining tasks exist, such as securing a joint using a pin or glue; using tape to attach two pieces of paper together; or, as we focus on in this work, building a simple electrical circuit consisting of a battery pack, a light emitting diode (LED), wires, and conductive putty.

The primary contribution of our work is takeaways and design desiderata intended to support the use of virtual reality in research on human-robot interaction. We compare VR vs. real-world performance of the same task, and explore common failures in simulating HRI tasks in VR more generally. Although our study is comparatively small, it addresses under-studied questions on the role of VR in human-robot collaboration, and in making these comparisons, we attempt to draw lessons on using virtual reality settings to support such research more broadly.

2 THE RIVR SIMULATION ENVIRONMENT

RIVR [5] (Robot Interaction in Virtual Reality) is a VR simulation environment designed to allow human-robot interaction studies to be performed in VR in a way that is consistent with performing the same studies on a physical robot. The three primary components are the VR client, render server, and a ROS (Robot Operating System) server; ROS is a set of software libraries and tools used for building and managing robot applications, and is the primary component responsible for handling the simulated robot in the environment. RIVR was built using the Unity game engine and ROS, with ROS# providing the interface between them. RIVR is intended to work in "off the shelf" such that configuration is minimal. With RIVR, several challenges encountered in this study are possible to rectify given that the unity game engine can be programmed to mimic parts of reality's physics.

3 EXPERIMENT

In this task, a robot and human jointly build a circuit using a battery pack containing three AA batteries, alligator clips, two molded blocks of conductive putty, and a light emitting diode (LED). The robot prompts the participant to hand the LED to the robot, then instructs the participant on how to wire the battery pack to the conductive putty blocks. Once everything is wired together, the robot prompts the participant to present the blocks of conductive putty to the robot. The robot then attempts to place the LED bridging

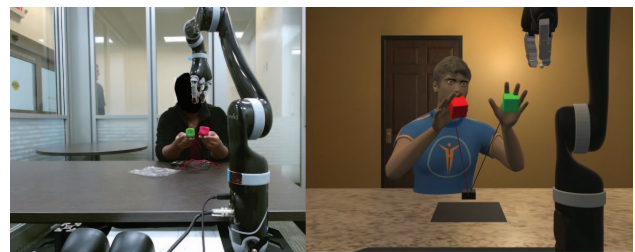


Figure 1: Participants interacting with the physical and virtual robot to build a simple electric circuit.

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This material is based in part upon work supported by the National Science Foundation under Grant Nos. 2024878, 1920079, and 2145642.

the blocks of putty, potentially retrying if problems are encountered. Once the trial is either successful or the user felt that it was not going to be successful, the interaction was concluded, and a short survey was given. This process was done both with a physical robot in the real world and with a virtual robot in a virtual reality simulation.

4 EXPERIMENTAL RESULTS

For this study twenty participants were recruited from a common area of a university campus. All were aged between 18 and 34, nine female and eleven male. Seven identified as white, five as Black or African American, four as Asian, two as Asian and white, and one as Black or African American and other. Ten of the participants interacted only with the virtual robot and the other ten only interacted with the physical robot.

Joining Task Effectiveness Despite the fact that the VR environment was developed in a simulator that is specifically intended to closely mirror human-robot interaction in the physical world, we found substantial differences in performance between the two settings, speaking to the difficulty of bridging the sim2real gap generally, and specifically to the complexities of conducting human-robot interaction work in simulation. While seven of the ten participants that worked with the virtual robot were able to successfully compete the task, only two were successful when working with the physical robot. In the next section we discuss sources of error in both environments, motivating our suggestions for improving on virtual reality based human-robot research.

Participant Experience Between Settings After the participants finished the experiment they were asked to evaluate how well they understood the instructions given by the robot, how useful they thought the task was, how comfortable they felt interacting with the robot, how frustrating the interaction was, how intuitive they found the interaction, and overall how pleasant they found the interaction using a five-point Likert scale. A one factor analysis of variance was performed to determine if there were significant differences in responses for virtual and real conditions. There were only significant differences in how comfortable participants were, with participants interacting with the physical robot feeling less comfortable (p-value of 0.035). There were not significant differences in understanding of the instructions, usefulness of the task, frustration, how intuitive they found the interaction, and how pleasant the interaction was.

We hypothesize that the significant difference in the level of comfort is primarily due to the difference in performance—that is, that users found it awkward to try to collaborate while the robot continually failed at its task. However, it is promising that the other responses did not show a significant difference, particularly in the intuitiveness of the interaction, showing that use of VR did not add any significant overhead to a participant's workload.

5 LESSONS AND CONCLUSIONS

The primary purpose of undertaking the study described in this paper was to better understand the role virtual reality can play in research into co-manipulation tasks and other shared physical tasks between a person and a robot. Even with the advent of several VR HRI open source projects, the development of a simulation of a particular VR simulation of an HRI task requires significant time and care. While this overhead should not be underestimated, it will often be substantially less than the cost of acquiring and maintaining a physical robot and a physical test environment. We suggest the following key takeaways:

1. In shared manipulation tasks, the participant's use of the VR controllers is extensive. It is therefore appropriate to consider the specific use of the controllers that is expected and perform early evaluations of the ergonomics and effectiveness of any selected control schema.

2. In physical human-robot interactions, the haptic feedback provided by, e.g., pulling pressure on a shared object is an important source of information for the person. Commodity VR hardware lacks the capacity to emulate physical resistance or obstruction, so this is a fundamental limitation of current VR technology and should be anticipated in the design process.
3. A robot's performance in the physical world is often constrained by the limitations of its sensors, so realistic emulation of noise is important in building simulations where a virtual robot will be faced with similar obfuscation and thus behave similarly to its real-world analog.
4. Although it is not necessary for every element of a virtual setting to closely mimic the physical reality of a robot's environment, what specific aspects are important to reproduce with high fidelity should be carefully considered and explicitly explored as part of the initial design process. Accurately modeling physics greatly improves the sense of realism, and collisions between manipulated objects should be modeled whenever it is possible.
5. Work that relies on soft-body interactions and other poorly supported physical models should be approached with caution, even if no significant object deformation is necessary to the completion of the task in question. During exploration of such tasks, it is worthwhile to consider the engineering tradeoffs of implementing such models in the virtual reality model.

Despite these difficulties, we have presented a small-scale initial study of successfully using virtual reality to study and understand human-robot interactions in a physically collaborative task. This study was performed in similar but non-identical virtual and physical settings. While not every element of these experiences will be informative for every case, we hope that some subset of these design suggestions has value for others working in similar areas. The primary contribution of this work is a set of lessons learned and design desiderata suitable for taking under consideration when leveraging virtual reality to extend the availability and effectiveness of human-robot interaction studies.

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